

IEM KOLKATA
Calculus
Limits & Continuity

Q1) $f(x) = |x| = \begin{cases} -x & x < 0 \\ x & x \geq 0 \end{cases}$

- (a) continuous only at $x = 0$
 (b) discontinuous at $x = 0$
 (c) continuous $\forall x$
 (d) None

Q2) $f(x) = \begin{cases} \frac{1}{1+2^{1/x}} & x \neq 0 \\ 0 & x = 0 \end{cases}$

Check if the function is continuous at $x = 0$.

Q3) $\lim_{x \rightarrow 0} \frac{1}{x}$

- (a) 0 (b) ∞ (c) $-\infty$ (d) does not exist

Q4) $f(x) = \begin{cases} 2x + 1 & x \leq 1 \\ ax^2 + b & 1 < x < 3 \\ 5x + 2a & x \geq 3 \end{cases}$

This function is continuous $\forall x$. Find a, b .

Q5) $\lim_{x \rightarrow 0} \frac{\tan 3x}{\sin 5x} = ?$

Q6) $\lim_{n \rightarrow \infty} \frac{n}{\sqrt{n^2 + n}} = ?$

Q7) $\lim_{x \rightarrow 0} x \sin\left(\frac{1}{x}\right) = ?$

Q8) $\lim_{x \rightarrow \infty} \frac{x - \sin x}{x + \cos x} = ?$

Q9) $\lim_{x \rightarrow 8} \frac{x^{\frac{1}{3}} - 2}{x - 8} = ?$

Q10) $\lim_{x \rightarrow 0} \frac{e^x - \left(1 + x + \frac{x^2}{2}\right)}{x^3} = ?$

Q11) $\lim_{x \rightarrow \infty} \left(\frac{1+x}{1+2x}\right)^x = ?$

Q12) $\lim_{x \rightarrow \infty} \left(\frac{5-x}{3-x}\right)^x = ?$

Q13) $\lim_{n \rightarrow \infty} n^{\frac{1}{n}} = ?$

Q14) $\lim_{n \rightarrow \infty} \left(1 - \frac{1}{n}\right)^{2n} = ?$

Q15) $\lim_{x \rightarrow 0} \frac{\tan x - x}{x^3} = ?$

Q16) $\lim_{x \rightarrow 0} \left(\frac{1}{x^2} - \frac{1}{\sin^2 x}\right) = ?$

Q17) $\lim_{x \rightarrow 0} \frac{|x|}{x} = ?$

Q18) $\lim_{x \rightarrow 0} e^x (\cos x)^{\frac{1}{\sin^2 x}} = ?$

Q19) $\lim_{x \rightarrow 0} \frac{2^{8 \cos x} [\sin^8(\frac{\pi}{8} + x) - \sin^8 \frac{\pi}{6}]}{8x} = ?$

Q20) Which of the following functions is continuous at $x = 3$

A. $f(x) = \begin{cases} 2 & \text{if } x = 3 \\ x - 1 & \text{if } x > 3 \\ \frac{x+3}{3} & \text{if } x < 3 \end{cases}$

B. $f(x) = \begin{cases} 4 & \text{if } x = 3 \\ 8 - x & \text{if } x \neq 3 \end{cases}$

C. $f(x) = \begin{cases} x + 3 & \text{if } x \leq 3 \\ x - 4 & \text{if } x > 3 \end{cases}$

D. $f(x) = \frac{1}{x^3 - 27} \text{ if } x \neq 3$